AI-Based Crop Disease Detection System

A PROJECT REPORT

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Under the guidance of,

Dr. Afroz Pasha

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence & Machine Learning).

At

PRESIDENCY UNIVERSITY

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PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

CERTIFICATE

This is to certify that the Project report **"AI-BASED CROP DISEASE DETECTION SYSTEM"** being submitted by **"TALLA SUNIL KUMAR, CHALLA YOGESH, SALAPAKSHI SAGAR, GADE PRATHYUSHA, JAMPANA VENKATA ARJUN VARMA"** bearing roll number(s) **"20211CAI0198, 20211CAI0162, 20211CAI0094, 20211CAI0200, 20211CAI0060"** in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence & Machine Learning) is a Bonafide work carried out under my supervision.

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DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **AI-BASED CROP DISEASE DETECTION SYSTEM** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence & Machine Learning)**, is a record of our own investigations carried under the guidance of **Dr. Afroz Pasha, Assistant professor Senior Scale, School of Computer Science Engineering, Presidency University, Bengaluru.**

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

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ABSTRACT

Agriculture plays a pivotal role in sustaining economies worldwide, providing food, employment, and raw materials for various industries. However, one of the most persistent challenges in the agricultural sector is crop diseases, which can lead to devastating yield losses and severely impact food security. Traditional methods of disease detection rely heavily on manual inspection and expert analysis, which are often time-consuming, expensive, and prone to human error. The inability to detect and diagnose plant diseases at an early stage can result in uncontrolled outbreaks, leading to significant economic losses for farmers and reduced crop productivity.

To address these challenges, this project introduces an AI-powered crop disease detection and diagnosis system, integrating state-of-the-art deep learning, natural language processing (NLP), and geospatial mapping technologies. The system utilizes YOLOv8 (You Only Look Once, version 8), a highly efficient Convolutional Neural Network (CNN) model, to perform real-time image-based detection of plant diseases. The model is trained on an extensive, well-annotated dataset from Roboflow, enabling it to classify and identify various diseases affecting rice, wheat, and maize crops with high accuracy.

In addition to automated disease detection, the system features an AI-driven chatbot based on Large Language Models (LLMs) to provide interactive diagnosis, treatment recommendations, and preventive measures. This chatbot acts as a virtual agricultural expert, offering farmers instant, context-aware advice on how to manage detected diseases, what pesticides or organic treatments to use, and how to prevent future outbreaks. The integration of NLP in the chatbot ensures that farmers can communicate in natural language, making the system accessible and easy to use even for individuals with limited technical expertise.

A key innovation in this project is the mapping feature, which leverages OpenStreetMap's Overpass API to identify nearby plant and pesticide shops based on the farmer's location. Once a disease is detected, farmers can instantly locate the nearest agricultural supply stores to purchase the required pesticides, fertilizers, or organic treatments. This geospatial integration bridges the gap between disease detection and resource accessibility, making it easier for farmers to take immediate corrective action.

The proposed system offers a holistic, AI-driven solution for crop disease management by combining image-based deep learning detection, chatbot-assisted diagnosis, and geolocation-based resource mapping. By reducing reliance on manual disease identification, improving treatment decision-making, and enhancing access to agricultural resources, this project contributes to a more efficient, technology-driven approach to modern farming. The implementation of real-time, AI-powered disease detection not only enhances agricultural productivity but also helps mitigate economic losses, ensuring a sustainable and resilient food production system.

Keywords: Crop Disease Detection, YOLOv8 CNN, Deep Learning in Agriculture, AI Chatbot, Large Language Models (LLMs), Plant Disease Classification, Geospatial Mapping, OpenStreetMap Overpass API, Precision Farming, AI-Powered Smart Agriculture

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